

Week 7 demo — fallback captures

Bike state machine: trace the lifecycle against the domain

Captured with Claude Code, Sonnet 5 (low effort), September
2026

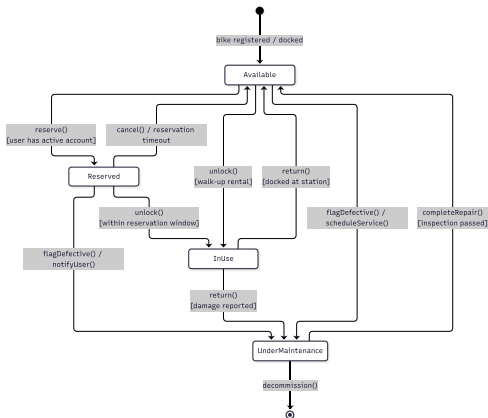
When to use this deck

Switch here if the live AI fails (no answer after 20 seconds, network down, garbage output). Walk the same defect catalogue against these captures: prompt #1, the revision after prompt #2, and the edge-case reserve.

Prompt #1

“Generate a UML state machine diagram (as Mermaid) for a bike in the city bike-sharing app: it can be available, reserved, in use, and under maintenance.”

AI's answer — the diagram



AI's answer — what it said

Notes

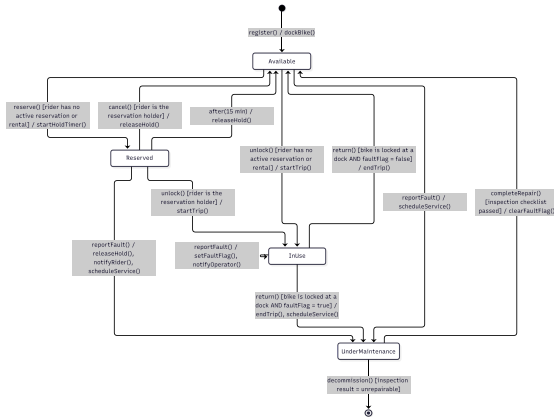
- ▶ **Available:** docked and rentable.
- ▶ **Reserved:** held for one user for a limited time. It returns to Available on cancel or timeout.
- ▶ **InUse:** rented and unlocked. It can only leave through a return at a station.
- ▶ **UnderMaintenance:** out of service. It can be entered from any state when a defect is found, and it is left only after repair and inspection (or decommissioning).

I chose to send a bike whose defect is reported mid-ride to maintenance on return, not mid-trip. If you want a different rule, such as a “Lost/Stolen” state, tell me and I’ll add it.

Prompt #2

“Revise the state machine. A rider can also unlock an available bike without reserving it. A fault can be reported during a ride, and a reserved bike can be flagged as faulty before pickup. Replace vague guards with conditions a tester could check, and use one transition per event — event / action means an action, not an alternative event. Keep an initial and a final state.”

AI's revision — the diagram



Al's revision — what it said I

What changed

- ▶ **Walk-up rental:** Available $\rightarrow$ InUse on unlock() is now explicit. It is a separate transition from the reserved unlock, and each has its own guard.
- ▶ **Fault during a ride:** reportFault() in InUse is a self-transition that sets faultFlag. The rider is not stranded mid-trip. The two return() transitions are mutually exclusive on faultFlag, so the bike goes to maintenance only once it is docked.
- ▶ **Fault on a reserved bike:** Reserved $\rightarrow$ UnderMaintenance on reportFault() releases the hold and notifies the rider.
- ▶ **One event per transition:** the old cancel() / reservation timeout is now two transitions. cancel() is the rider's action, and after(15 min) is a time event. The text after / lists only actions.

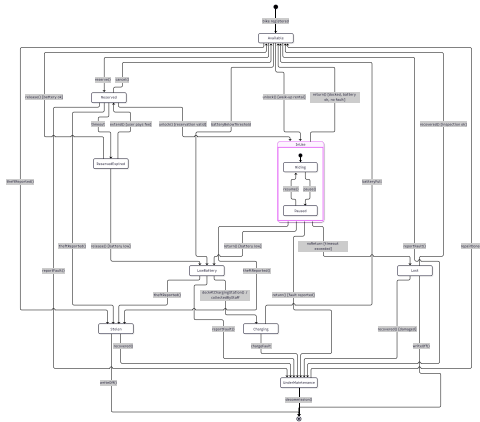
AI's revision — what it said II

- ▶ **Testable guards:** each guard can be checked from system data, such as the rider's active bookings, the reservation holder, dock lock status, `faultFlag`, or the inspection result. The 15-minute hold is an assumed value, so adjust it to your policy.
- ▶ **Initial and final states:** the initial transition is `register()`, and the final one is `decommission()`.

Reserve prompt — grow the state space

“Add all the edge-case states: lost, stolen, reserved-but-expired, charging, low-battery. Model ‘in use’ as a composite state with Riding and Paused substates.”

Reserve — the diagram



Reserve — what it said I

Design decisions:

- ▶ **InUse** is a composite state: entering starts in **Riding**, and **Paused** is a temporary stop, such as a lock during a shop visit. Transitions on the outer InUse (return, lost, stolen, low battery, fault) apply from either substate.
- ▶ **ReservedExpired** is a holding state. The reservation timed out, and the bike is released to Available or LowBattery depending on its charge. The user can also pay to extend.
- ▶ **LowBattery** blocks rental. The bike goes to Charging after a dock or staff pickup.
- ▶ **Lost** means a ride was never ended. **Stolen** means an external theft report. Both can be recovered, after inspection or repair, or written off.
- ▶ Recovered stolen bikes always go to maintenance, since their condition is unknown.

Assumptions you may want to change:

Reserve — what it said II

- ▶ Mermaid can't draw a transition from a substate to an outer state cleanly, so `return()` leaves from the composite boundary.
- ▶ I omitted Available → Charging for opportunistic top-ups. It's easy to add.